

class 14: RNASeq Mini-Project

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Background

The data for today's mini-project comes from a knock-down study of an important HOX gene.

Data Import

```
countFile <- "GSE37704_featurecounts.csv"
metaFile  <- "GSE37704_metadata.csv"

colData    <- read.csv(metaFile, row.names = 1)
countData  <- read.csv(countFile, row.names = 1)
```

Have a wee peak at these:

```
head(colData)
```

```
          condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
head(countData)
```

```
          length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
          SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

###Clean up (data tidying)

We need to remove the funny “length” column from our `countData` to make the columns match the rows in `colData`.

```
countData <- countData[,-1]
```

Check match of `colData` and `countData`

```
rownames(colData) == colnames(countData)
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE
```

```
###Remove zero count genes
```

```
to.keep <- rowSums(countData) >0  
countData <- countData[to.keep,]
```

DESeq Analysis

```
library(DESeq2)
```

Setting up the DESeq object

```
dds <- DESeqDataSetFromMatrix(countData = countData,  
                              colData   = colData,  
                              design    = ~ condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

Running DESeq

```
dds= DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

Getting results

```
res <- results(dds)
res
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 15975 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
...	...	...	...	...	...
ENSG00000273748	35.30265	0.674387	0.303666	2.220817	2.63633e-02
ENSG00000278817	2.42302	-0.388988	1.130394	-0.344117	7.30758e-01
ENSG00000278384	1.10180	0.332991	1.660261	0.200565	8.41039e-01
ENSG00000276345	73.64496	-0.356181	0.207716	-1.714752	8.63908e-02
ENSG00000271254	181.59590	-0.609667	0.141320	-4.314071	1.60276e-05
	padj				
	<numeric>				
ENSG00000279457	6.86555e-01				
ENSG00000187634	5.15718e-03				
ENSG00000188976	1.76549e-35				
ENSG00000187961	1.13413e-07				
ENSG00000187583	9.19031e-01				
...	...				
ENSG00000273748	4.79091e-02				
ENSG00000278817	8.09772e-01				
ENSG00000278384	8.92654e-01				
ENSG00000276345	1.39762e-01				
ENSG00000271254	4.53648e-05				

Volcano Plot

```
library(ggplot2)

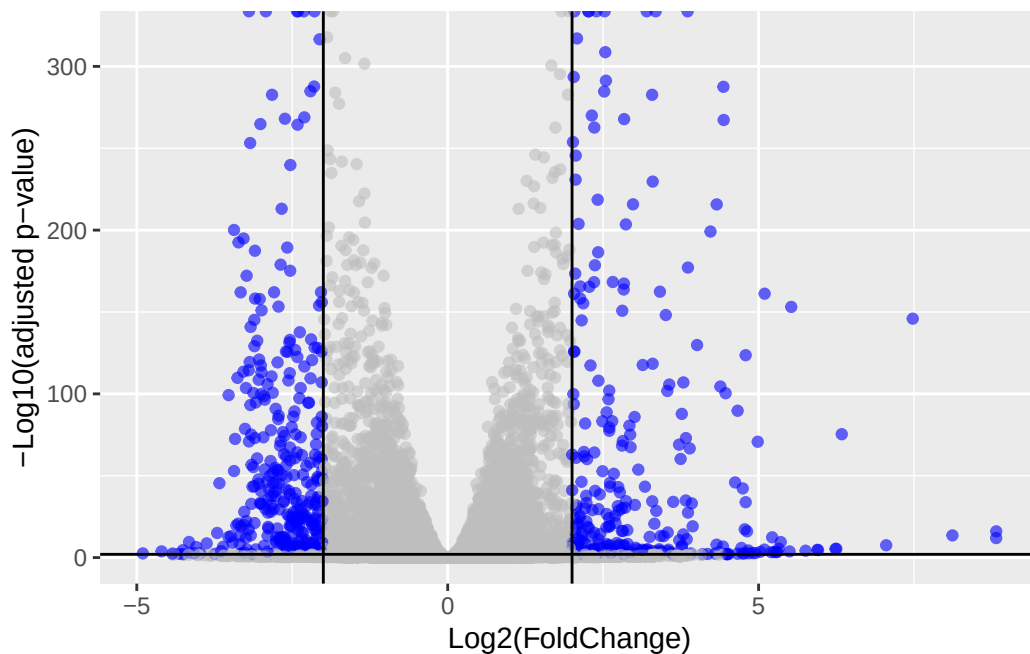
mycols <- rep("gray", nrow(res))
```

```
mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"

mycols[ res$padj > 0.01 ] <- "gray"

ggplot(as.data.frame(res)) +
  aes(log2FoldChange,
      -log10(padj)) +
  geom_point(col = mycols, alpha = 0.6) +
  xlab("Log2(FoldChange)") +
  ylab("-Log10(adjusted p-value)") +
  geom_vline(xintercept = c(-2, 2)) +
  geom_hline(yintercept = -log10(0.01))
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



##Add Annotation

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys      = row.names(res),
                     keytype   = "ENSEMBL",
                     column    = "SYMBOL",
                     multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys      = row.names(res),
                     keytype   = "ENSEMBL",
                     column    = "ENTREZID",
                     multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name <- mapIds(org.Hs.eg.db,
                   keys      = row.names(res),
                   keytype   = "ENSEMBL",
                   column    = "GENENAME",
                   multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.913579	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.188076	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.922128	2.0570638	0.1969053	10.446970	1.51282e-25
ENSG00000187608	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02

ENSG00000188157	9128.439422	0.3899088	0.0467163	8.346304	7.04321e-17
ENSG00000237330	0.158192	0.7859552	4.0804729	0.192614	8.47261e-01
	padj	symbol	entrez		name
	<numeric>	<character>	<character>		<character>
ENSG00000279457	6.86555e-01	NA	NA		NA
ENSG00000187634	5.15718e-03	SAMD11	148398	sterile alpha motif ..	
ENSG00000188976	1.76549e-35	NOC2L	26155	NOC2 like nucleolar ..	
ENSG00000187961	1.13413e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.19031e-01	PLEKHN1	84069	pleckstrin homology ..	
ENSG00000187642	4.03379e-01	PERM1	84808	PPARGC1 and ESRR ind..	
ENSG00000188290	1.30538e-24	HES4	57801	hes family bHLH tran..	
ENSG00000187608	2.37452e-02	ISG15	9636	ISG15 ubiquitin like..	
ENSG00000188157	4.21963e-16	AGRN	375790		agrin
ENSG00000237330	NA	RNF223	401934	ring finger protein ..	

```
res <- res[order(res$pvalue), ]
write.csv(as.data.frame(res), file = "deseq_results.csv")
```

Pathway Analysis

```
library(pathview)
library(gage)
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

kegg.sets.hs <- kegg.sets.hs[sigmet.idx.hs]
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
```

```
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"
```

KEGG

```
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez

foldchanges <- foldchanges[!is.na(names(foldchanges))]

head(foldchanges)
```

```
1266 54855 1465 2034 2150 6659
-2.422719 3.201955 -2.313738 -1.888019 3.344508 2.392288
```

```
keggres <- gage(foldchanges, gsets = kegg.sets.hs)

head(keggres$greater)
```


	p.geomean	stat.mean	p.val
hsa04640 Hematopoietic cell lineage	0.002768235	2.840168	0.002768235
hsa04630 Jak-STAT signaling pathway	0.005047208	2.596424	0.005047208
hsa00140 Steroid hormone biosynthesis	0.007156705	2.532159	0.007156705
hsa04142 Lysosome	0.009770966	2.351292	0.009770966
hsa04330 Notch signaling pathway	0.018381844	2.120084	0.018381844
hsa04916 Melanogenesis	0.018931153	2.092166	0.018931153

	q.val	set.size	exp1
hsa04640 Hematopoietic cell lineage	0.3840765	55	0.002768235
hsa04630 Jak-STAT signaling pathway	0.3840765	109	0.005047208
hsa00140 Steroid hormone biosynthesis	0.3840765	31	0.007156705
hsa04142 Lysosome	0.3932814	118	0.009770966
hsa04330 Notch signaling pathway	0.4331850	46	0.018381844
hsa04916 Melanogenesis	0.4331850	90	0.018931153

```
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	9.366328e-06	-4.369134	9.366328e-06
hsa03030 DNA replication	9.592718e-05	-3.946505	9.592718e-05
hsa03013 RNA transport	1.439613e-03	-3.014203	1.439613e-03
hsa03440 Homologous recombination	3.106306e-03	-2.848189	3.106306e-03
hsa04114 Oocyte meiosis	3.888649e-03	-2.688749	3.888649e-03
hsa00010 Glycolysis / Gluconeogenesis	9.125398e-03	-2.398384	9.125398e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001507979	121	9.366328e-06
hsa03030 DNA replication	0.007722138	36	9.592718e-05
hsa03013 RNA transport	0.077259252	144	1.439613e-03
hsa03440 Homologous recombination	0.125028834	28	3.106306e-03
hsa04114 Oocyte meiosis	0.125214484	102	3.888649e-03
hsa00010 Glycolysis / Gluconeogenesis	0.213990959	53	9.125398e-03

```
keggrespathways_up <- rownames(keggres$greater)[1:5]
keggresids_up <- substr(keggrespathways_up, 1, 8)

pathview(gene.data = foldchanges, pathway.id = keggresids_up, species = "hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/user/Desktop/bimm 143/class14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/user/Desktop/bimm 143/class14

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/user/Desktop/bimm 143/class14

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/user/Desktop/bimm 143/class14

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/user/Desktop/bimm 143/class14

Info: Writing image file hsa04330.pathview.png

```
keggrespathways_down <- rownames(keggres$less)[1:5]
keggresids_down <- substr(keggrespathways_down, 1, 8)
pathview(gene.data = foldchanges, pathway.id = keggresids_down, species = "hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/user/Desktop/bimm 143/class14

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/user/Desktop/bimm 143/class14

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/user/Desktop/bimm 143/class14

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/user/Desktop/bimm 143/class14

Info: Writing image file hsa03440.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/user/Desktop/bimm 143/class14

Info: Writing image file hsa04114.pathview.png

G0:0007156	homophilic cell adhesion	8.132355e-05	3.836423	8.132355e-05
G0:0002009	morphogenesis of an epithelium	1.301491e-04	3.672346	1.301491e-04
G0:0048729	tissue morphogenesis	1.324453e-04	3.663702	1.324453e-04
G0:0007610	behavior	1.786592e-04	3.585306	1.786592e-04
G0:0035295	tube development	5.558494e-04	3.273391	5.558494e-04
G0:0060562	epithelial tube morphogenesis	5.613074e-04	3.277405	5.613074e-04

		q.val	set.size	exp1
G0:0007156	homophilic cell adhesion	0.1804788	113	8.132355e-05
G0:0002009	morphogenesis of an epithelium	0.1804788	339	1.301491e-04
G0:0048729	tissue morphogenesis	0.1804788	424	1.324453e-04
G0:0007610	behavior	0.1825897	426	1.786592e-04
G0:0035295	tube development	0.3325708	391	5.558494e-04
G0:0060562	epithelial tube morphogenesis	0.3325708	257	5.613074e-04

\$less

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.735928e-15	-8.047667	1.735928e-15
G0:0000280	nuclear division	4.808128e-15	-7.923688	4.808128e-15
G0:0007067	mitosis	4.808128e-15	-7.923688	4.808128e-15
G0:0000087	M phase of mitotic cell cycle	1.312717e-14	-7.781713	1.312717e-14
G0:0007059	chromosome segregation	2.149699e-11	-6.868718	2.149699e-11
G0:0000236	mitotic prometaphase	1.794819e-10	-6.689146	1.794819e-10

		q.val	set.size	exp1
G0:0048285	organelle fission	6.551876e-12	376	1.735928e-15
G0:0000280	nuclear division	6.551876e-12	352	4.808128e-15
G0:0007067	mitosis	6.551876e-12	352	4.808128e-15
G0:0000087	M phase of mitotic cell cycle	1.341596e-11	362	1.312717e-14
G0:0007059	chromosome segregation	1.757594e-08	142	2.149699e-11
G0:0000236	mitotic prometaphase	1.222870e-07	84	1.794819e-10

\$stats

		stat.mean	exp1
G0:0007156	homophilic cell adhesion	3.836423	3.836423
G0:0002009	morphogenesis of an epithelium	3.672346	3.672346
G0:0048729	tissue morphogenesis	3.663702	3.663702
G0:0007610	behavior	3.585306	3.585306
G0:0035295	tube development	3.273391	3.273391
G0:0060562	epithelial tube morphogenesis	3.277405	3.277405

Reactome

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]  
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes,  
            file = "significant_genes.txt",  
            row.names = FALSE,  
            col.names = FALSE,  
            quote = FALSE)
```

When you mine Reactome, the leading pathway is typically a cell-cycle/mitosis-related pathway for this HOXA1 knockdown dataset, which is as expected from the KEGG result where Cell cycle (hsa04110) is significantly enriched for downregulated genes. Discrepancies between Reactome and KEGG can occur because they employ distinct pathway definitions and curation, distinct gene membership, and your analysis relies on ID mapping completeness (Ensembl to Entrez/Symbol) and the statistical analysis (gag tests coordinated shifts in a ranked fold-change vector rather than simple “significant gene list” over-representation).